

Retain the body; investigate local repairs

Whole-body QA recommendation. This is an unaccepted finding, not asset or player approval.

Retain the frozen 13,380-vertex fitting body as the anatomical base. Its rest surface is closed, connected and consistently wound. Motion produces shoulder, hip and knee folds in both full-native and normalized-four fields. The evidence supports a bounded local deformation investigation before any body replacement.

What the measurements distinguish

Domain	Finding / disposition
Rest topology	One component; 26,756 triangles; Euler 2. No boundary, non-manifold edge, inconsistent edge winding, zero-area face or detected non-adjacent self-contact.
Pose / weights	Synthetic peak: 220 strict pairs. Actual47 peak: 414. Finite contact totals match at every pose. Strict counts differ by two at one tick; a fifth slot alone cannot resolve these folds.
Export consumption	Appearance10 diagnostic body maps all 9,037 native vertices and 18,016 oriented triangles. Primary-four field matches existing native-four stream within 0.669 micrometre.
Protected head	Assess separately. Native head seams join virtually to one component / six boundary loops. Neck interface overlap is deliberate; actual native head skin also develops self-crossings. Head oriented export ancestry remains unproven.
Visual decision	Existing continuous films are dressed. Exposed shoulder/hip quality, target garment19 export, facial behavior and physical iOS remain unreviewed. Parent judges art.

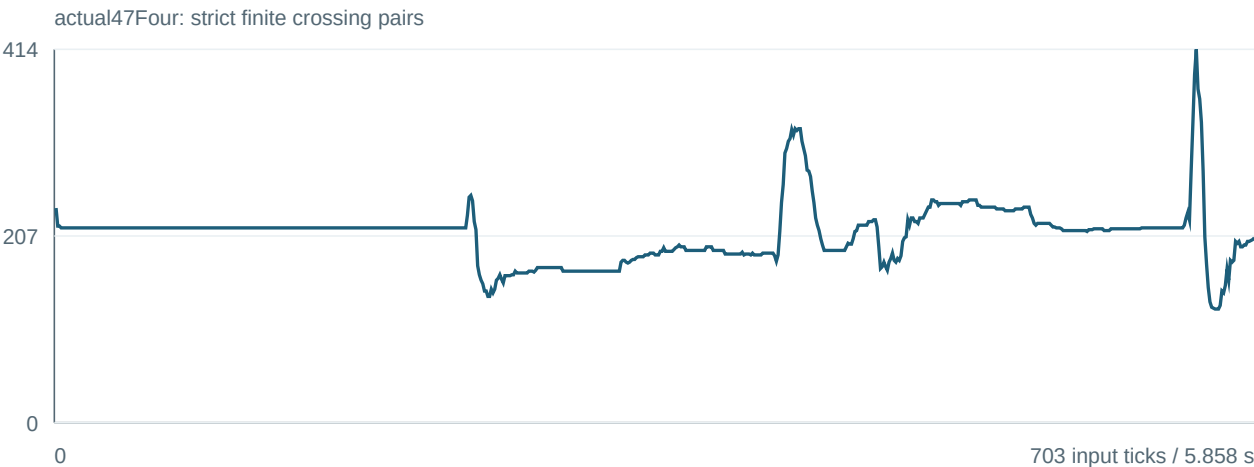
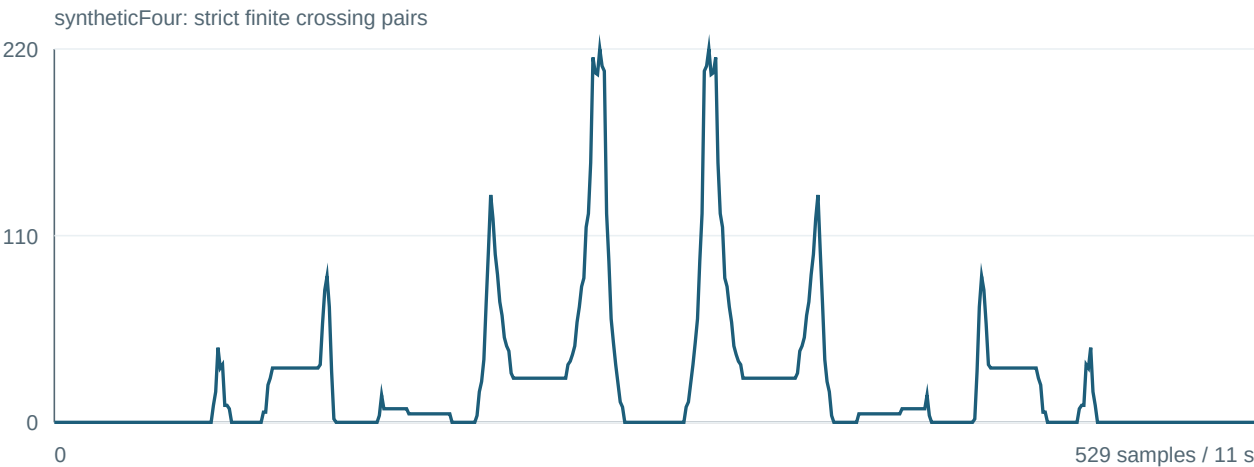
Shoulder and hip proxies

Rest spheres have radius 145.8 mm (8% of 1.82257 m figure height), centered on upper-arm and thigh bone heads. These overlap nearby torso surfaces; they are not anatomical segmentation.

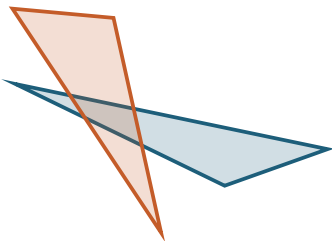
Region	Synthetic strict peak	Actual47 strict peak	Four-slot loss maximum
Shoulder L / R	22 / 22	69 / 69	0.261 mm synthetic / 0.357 mm actual
Hip L / R	35 / 40	49 / 53	0.000137 mm synthetic / 0.00000455 mm actual

Contacts arise under motion

Counts are sampled finite triangle crossings; they are not penetration depth or visible severity.



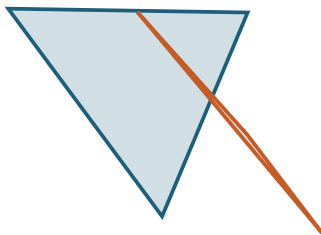
Synthetic Full and Four: 274 / 529 samples have strict crossings, with a 220-pair peak at frame 240 (5.0 s). Full/four maxima differ by 7.811 mm at native vertex 931 near the shoulder-neck blend. Actual47 full/four maximum is 6.012 mm. At actual input 478, both fields have 223 finite contacts; strict counts are Full 221 / Four 223. Regional strict counts match at every native/actual47 sample.



Isometric projection of existing numeric witness

Shoulder L, input 668
Triangles 2275 / 7130; chest / upper arm. Strict plane endpoint excursion: 15.105 mm.

2026-10-04 | Frozen evidence only | All M0-M5 open

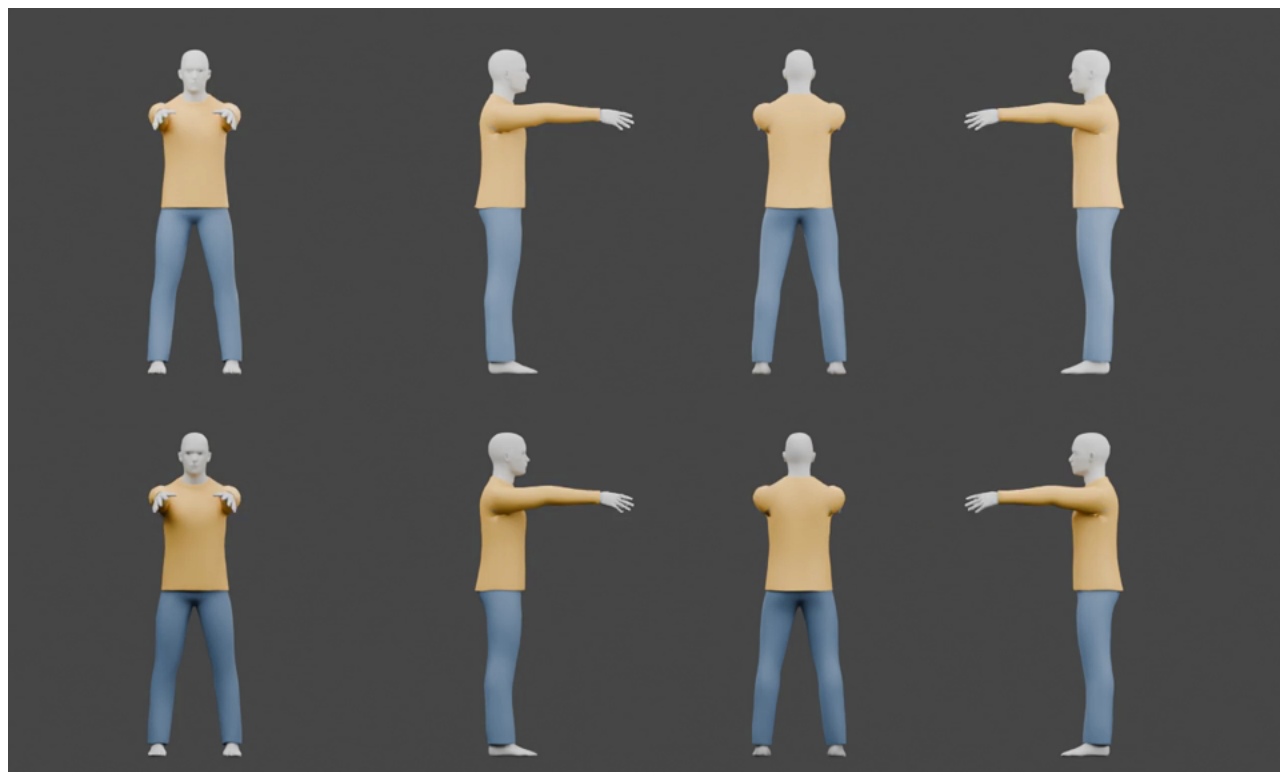


Isometric projection of existing numeric witness

Hip L, input 512
Triangles 8217 / 21263; thigh / pelvis. Strict plane endpoint excursion: 12.750 mm.

Played evidence has a visibility limit

Existing movies replayed silently in headless WebKit; these are extracts, not new posed captures.



Native stream, 2.0 s / frame 96 - TOP original full-native weights; BOTTOM normalized four. Four yaw columns. Existing shirt and jeans conceal the exposed body surface.



Native stream, 5.0 s / frame 240 - Existing backward-seated FK endpoint, within an uninterrupted forward/reverse movie. This visual evidence does not prove naked shoulder or hip clearance.

Provenance and remaining coverage

No body/head/rig/weights were edited. No new scene capture, export, installation or model job ran.



Separate actual50 OFF / ON film, 3.0 s. Its 176 existing OFF bone-world samples also show a 414-pair body peak. Actual47 and actual50 poses differ (matrix component gap 0.293341); shared body attributes and inverse binds are byte-exact. Do not pair film50 frames with numeric47 witnesses. Cause is not established.

Method and scope

Canonical rest: identity pose basis on all 51 native bones; no action/NLA; default zero shape values. File-world glTF metres: +X forward, +Y up, +Z left. Blender +X / +Z / -Y converts once; X+0.65 is baked once. GLTFLoader uses identity bind; attached mesh-world cancellation prevents a second translation.

CPU BVH broadphase; normalized finite triangle SAT epsilon 1e-9 m. Vertex-sharing adjacency excluded. Strict crossing requires >2 micrometre opposite endpoint distances and interior barycentrics >1e-6. Fixed rest tessellation. 32 analytic checks and 36,632 independent JS witness checks pass. No degenerate candidate pair was treated as clear. Head uses native fields; complete oriented head export ancestry is not proven.

Coverage: native 529 samples / 48 Hz / 11 s, plus existing actual47 703 ticks / 120 Hz / 5.858 s and separate actual50 176 samples. One Rookie backward-lean window; no arbitrary pose, full 86-bank, Pro, landing, face, stranger or physical-device clearance. 68 strict head/body rest-interface overlaps are separate from body self-contact. Existing underwear is a fitting control; no fresh exposed-body film was generated.

Exact source hashes (full registry: [body52/source.json](#))

Native19 profile-fit.blend

b545229e52c4fc5bcb97b4fbc81ccc3a0e5a2f9c87fe3c7338691317d965e449

Existing appearance10 GLB

8e4c5cff0e35b00a7ebc1686e8494fde9ef6388e012c8eebda31a508b3bc343c

Appearance10 controller

c6ba4edd2fe80829760802a3526e6a2f8166314baefd6b5350bd94222c5c4bc3

Native pose driver

8c40c42af362a8d3d8ae44d27a0b07570f600b26adc66d325239413f66c7625b

Native full stream

e2ac607041fdde967c58ac020dca3d224048424e81ba218d2b6ca9bddfdb0cea

Native four stream

8a68b1a371197bdc5eb6267bc748b02adf213a100debf5d92dde37f9c66444a5